TORONTO REAL ESTATE MARKET

Data-Driven Intelligence

Transforming housing market pricing into transparent, actionable insights for buyers, sellers, and real estate professionals.

INTRODUCTION & PROBLEM OVERVIEW

Market Overview

- Toronto has one of the largest real estate markets in Canada
- Wide variety of properties (condos, townhouses, detached homes)
- Prices vary significantly across different locations

The Problem

- Prices vary depending on location, size, and property type
- Similar properties have varying prices
- Hard to know if a property is fairly priced

Our Approach (BI)

- Use BI to analyze Toronto housing data in Toronto
- Identify pricing patterns and trends
- Compare properties more effectively
- Support better, informed decision-making

Goal: Analyze Toronto housing data to understand pricing patterns and support better decisions for buyers, sellers, and realtors.

BI SOLUTION – 5 STAGE PIPELINE

1 Data Acquisition

Kaggle MLS Dataset

6,616 Toronto listings

396 raw columns

14 core columns selected

Bank of Canada API

Valet API · Series V39079

Overnight rate history

Python (Google Colab)

→

2 Data Cleaning

- Removed duplicate rows
- Standardized column names
- Dropped square footage (97% missing)
- Imputed property tax by locality median
- Filtered outliers

Pandas

→

3 Storage

housing_cleaned
Cleaned data written to SQLite

Rate data CSV
Bank of Canada interest rates

SQLite

→

4 Analysis

Location
FSA median price by first 3 postal chars

Market
Boxplot by property type (log scale)

Configuration
Bed/bath combo median prices

Feature
Price vs Property Tax scatter

Pandas · Seaborn · Matplotlib

→

5 Visualization

- Interactive Power BI dashboard
- 5 pages with dynamic filters
- Filter by type, price, neighborhood
- BoC rate trend as market context
- SQLite as data source for Power BI

Power BI

KEY FINDINGS

What the Data Revealed

01 LOCATION

M2L

Highest-priced FSA: ~$5M median

Top 10 FSAs span $2M–$5M median. Central & north zones command strong premiums over the city-wide median of $780K.

02 MARKET COMPOSITION

66.8%

of listings are condos

4,393	condos · median $660K
1,801	single-family · median $1.6M
117	condo/townhome · $799K

03 CONFIGURATION

Baths influence price.

Fix 3 beds — vary baths:

1 bath	**$644K**
2 baths	**$899K**
3 baths	**$1.30M**
4 baths	**$1.85M**

Fix 3 baths — vary beds: $1.3M–$1.4M (flat)

04 FEATURE DRIVERS

Tax r=0.74*

strongest correlation

Property tax shows the highest correlation with price. However, ~20% of tax values were imputed from locality medians, which may influence this value.

* ~20% of tax values imputed — treat as supporting indicator

POWER BI DASHBOARD

Interactive Market Overview

This dashboard provides an interactive overview of the Toronto housing market, allowing users to explore pricing patterns, location difference, and property characteristics through dynamic filters.

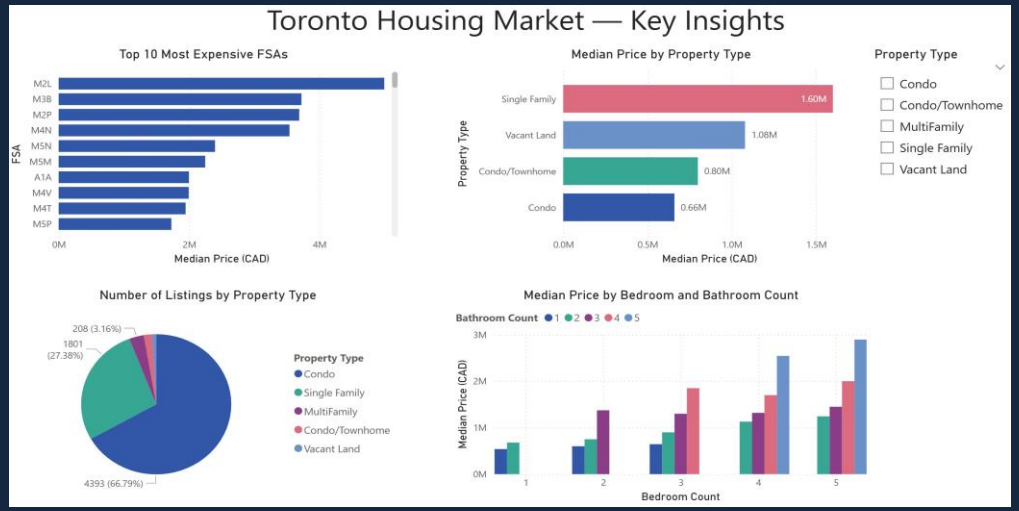
01 Summary
Overview of key insights across all analysis areas

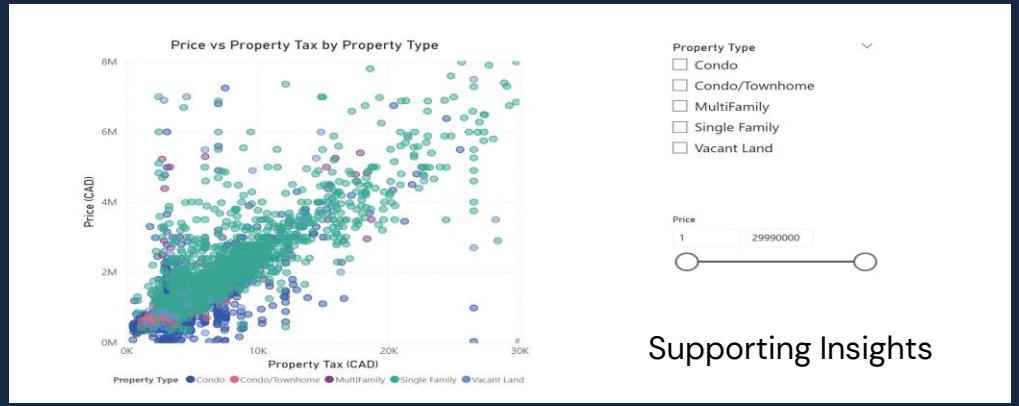
02 Market Overview
Price distribution and listing volume by property type

03 Location Analysis
Top 10 FSAs by median price and geographic price distribution

04 Property Configuration
Median price by bedroom and bathroom configuration

05 Feature Drivers
Relationship between property tax and price by property type





Supporting Insights

Tool: Power BI

BENEFITS

What BI Delivers

Five key advantages that transform Toronto's housing market into a transparent, data-driven ecosystem — helping buyers, sellers, agents, and investors.

Better Decisions Pricing by FSA, type & config in seconds — no manual research needed

Market Transparency Same data for all stakeholders; closes the buyer-seller information gap

Smart Valuation Bathrooms outperform bedrooms as a price predictor — a key insight for sellers

Time Savings Hours of manual research replaced by interactive Power BI filters

Macro Context Bank of Canada interest rate trends overlaid for long-term decisions

CHALLENGES & LIMITATIONS

What We Faced

01 DATA QUALITY

Corrupt property tax entries (e.g.,$888Merrors) required extensive filtering before analysis could begin.

02 GEOGRAPHIC INCONSISTENCY

Inconsistent neighborhood n ame formats required switching to FSA postal codes for reliable geographic grouping.

03 INTEREST RATE INTEGRATION

Missing consistent listing dates prevented direct joining of interest rate data with individual property records.

04 STATIC DATASET

Insights reflect a point-in-time snapshot; alive data pipeline would be needed for ongoing market monitoring.

Recommendation: "A live API pipeline would transform this into a real-time market intelligence platform."

CONCLUSION

From Opacity to Intelligence

This project demonstrates that BI methodologies can democratize Canada's largest real estate market — turning fragmented listings into clear, actionable insights for every stakeholder.

$90B annual Toronto market value

5 pipeline stages fully implemented

100K+ annual transactions analyzed